

Cognitive Workload Allocation Under Uncertainty

A Stochastic Multiple Knapsack Approach with Workload Balance

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Agenda

- 1 Motivation
- 2 Problem Formulation
- 3 Visual Intuition
- 4 Methodological Context
- 5 Benders Decomposition
- 6 Computational Results
- 7 Takeaways
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Motivation: Cognitive Overload, Burnout

- Cognitive overload from excessive demands leads to billions of lost working days annually [Robinson, 2024].
- Productivity depends on **attention management** rather than rigid time scheduling [Grant, 2019].
- Traditional time management fails because it ignores the finite nature of human lifespan and cognitive capacity [Hart, 2024].
- Chronic illness, burnout risk, and work-family conflict underscore the need for systematic resource allocation that accounts for well-being [Cook and Zill, 2024, Yoo and Lee, 2022].

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- A workload-balance penalty $\lambda \geq 0$ discourages uneven utilization across agents.

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Auxiliary variables

- $y_j^\omega \in \{0, 1\}$: violation indicator
- $U_j \geq 0$: expected utilization
- $Z \geq 0$: max utilization gap

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The violation indicator $y_j^\omega = 1$ marks scenario ω as an overload scenario for agent j :

$$\sum_{i \in \mathcal{I}} S_{ij}^\omega x_{ij} \leq C_j + M y_j^\omega, \quad \forall j \in \mathcal{J}, \forall \omega \in \mathbf{\omega}.$$

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The total violation probability for each agent must stay below α :

$$\sum_{\omega \in \mathbf{\omega}} p_\omega y_j^\omega \leq \alpha, \quad \forall j \in \mathcal{J}.$$

Workload Utilization and Balance

Expected workload utilization of agent j , normalized by capacity:

$$U_j = \frac{1}{C_j} \sum_{i \in \mathcal{I}} \sum_{\omega \in \mathbf{\Omega}} p_{\omega} S_{ij}^{\omega} x_{ij}, \quad \forall j \in \mathcal{J}.$$

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- A smaller Z means a more balanced allocation.
- Normalization by C_j allows fair comparison across agents with different capacities.

Full Formulation

$$\max_{x,y,U,Z} \underbrace{\sum_j \sum_i R_{ij} x_{ij}}_{\text{total return}} \quad (\text{Obj})$$

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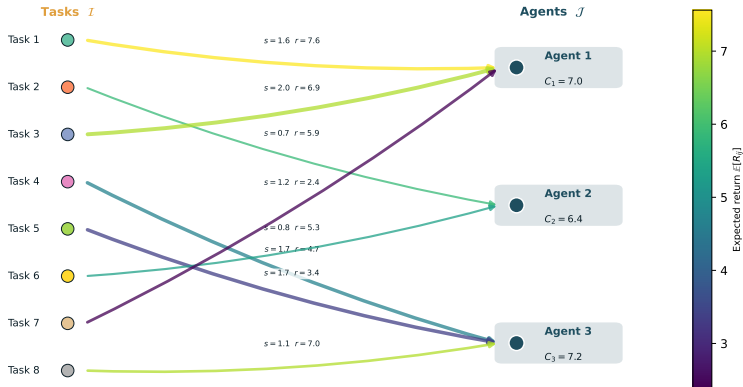
The objective balances three forces: return creation, overload-risk control, and workload fairness.

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Multiple Knapsack Structure

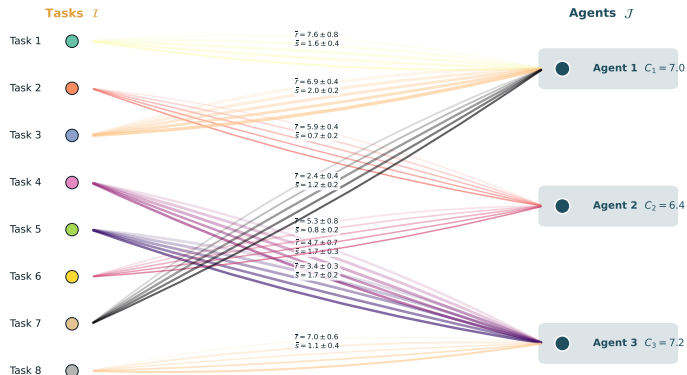
Multiple Knapsack: Bipartite Assignment Structure



Bipartite assignment: tasks (left) are assigned to agents (right) with capacities C_j . Edge width encodes task size $\mathbb{E}[S_{ij}]$ and color indicates expected return R_{ij} . Each task is assigned to at most one agent (constraint 10).

Stochastic Sizes and Returns

Stochastic MKP: Uncertain Sizes and Returns on Assignments



Stochastic extensions: each assignment edge shows multiple scenario realizations of size S_{ij}^ω (line width) and return R_{ij} (annotation). The chance constraint (10) limits the probability that total realized load exceeds agent capacity.

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Stochastic Knapsack: Cutting Planes and Dominance

- Büyüktaktın [2023] derive **scenario dominance cuts** for multi-stage **stochastic MIPs** applied to the dynamic multi-dimensional knapsack. Strong dominance cuts reduce solution time with 0.06% average deviation from optimality.
- Büyüktaktın [2022] generalize this via **Stage- t Scenario Dominance** for risk-averse (mean-CVaR) knapsack problems, achieving one-to-two orders of magnitude speedup.
- Joung et al. [2023] **compare LP relaxation bounds** for the robust knapsack problem and show that extended formulations yield identical bounds.

Distributionally Robust and Risk-averse Knapsack

- Bos et al. [2024] approximate the stochastic knapsack with periodic scheduled arrivals using **Wasserstein Distributed Robust Optimization (DRO)**, producing robust healthcare surgical schedules.
- Ding et al. [2022] propose a target-based DRO knapsack with **uncertain profit and capacity**, reformulated as a **second-order conic program**.
- Merzifonluoglu and Geunes [2021] study the **risk-averse static stochastic knapsack** with CVaR and mean-standard deviation objectives, deriving efficient heuristics for normally distributed resource consumption.
- Boonstra et al. [2025] solve a robust multi-item newsvendor with budget constraint using a greedy knapsack algorithm under moment ambiguity.

Solution Algorithms for Stochastic Programs

- Chen and Luedtke [2022] analyze **sample average approximation** for two-stage programs without complete recourse, proving exponential convergence of recourse likelihood.
- Larsen et al. [2024] accelerate L-shaped methods via **supervised ML**, solving stochastic multi-knapsack instances in 20% of the time with less than 0.1% optimality gap.
- Lefebvre et al. [2024] reformulate adjustable robust optimization with **discrete uncertainty** and apply **branch-and-cut to the multiple knapsack problem with uncertain weights**.
- Lyu et al. [2022] unify **responsive, adaptive, and anticipative allocation policies** for the stochastic knapsack through **persistence values**.

Assignment, Scheduling, and Workforce Planning

- Rezaeian et al. [2024] model the **stochastic assignment of projects** to managers via **multi-stage stochastic integer programming** with heuristic algorithms for large instances.
- Nasirian et al. [2025] solve multiskilled workforce staffing and scheduling via **logic-based Benders decomposition**, achieving 29x speedup over direct MILP solving.
- Cavagnini et al. [2020] use two-stage stochastic programming for production planning under **uncertain learning rates**, with insights on cross-training and worker assignment.
- Novak et al. [2022] connect stochastic parallel machine scheduling to an **inflatable stochastic knapsack** pricing subproblem within **branch-and-price**.

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- **Key insight:** the scenario-based chance constraints can be checked *after* a candidate assignment is fixed.
- Benders decomposition separates the problem into:
 - ▶ A **master problem** that decides assignments and evaluates the return-balance trade-off.
 - ▶ A **feasibility subproblem** that checks whether the candidate assignment satisfies chance constraints.

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- Benders decomposition separates the problem into:
 - ▶ A **master problem** that decides assignments and evaluates the return-balance trade-off.
 - ▶ A **feasibility subproblem** that checks whether the candidate assignment satisfies chance constraints.
- This mirrors the managerial logic: propose an assignment first, then verify safety under uncertainty.

General Benders Principle (from Lecture)

A large program $[P]$ with variables partitioned into groups x and y :

$$[P] \quad \min c'x + d'y \quad \text{s.t.} \quad Ax + By = e, \quad x \geq 0, \quad y \in Y.$$

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Step 3. Reformulate $[P]$ as a *master problem* with **optimality cuts** from extreme points and **feasibility cuts** from extreme rays of the dual domain.

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Constraint Generation

Instead of enumerating all cuts, we solve a relaxed master and add violated cuts iteratively. The algorithm converges in finitely many iterations because the number of extreme points and rays is finite.

Our Decomposition Structure

Notation. Define the expected workload coefficient:

$$\bar{S}_{ij} = \sum_{\omega \in \mathbf{\Omega}} p_{\omega} S_{ij}^{\omega}, \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J}.$$

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- The **workload-balance component** depends only on expected workload, so it stays in the master problem.
- The **scenario-based chance feasibility** is naturally evaluated only after a candidate assignment is fixed.

Master Problem (Iteration t)

Let $\mathcal{B}^t$ denote the set of feasibility cuts generated so far.

$$\max_{x, U, Z} \sum_j \sum_i R_{ij} x_{ij} - \lambda Z \quad (\text{MP-Obj})$$

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The master keeps the true objective and balance structure. Chance constraints are *not* included here; they are checked in the feasibility subproblem.

Feasibility Evaluation Subproblem

Given a candidate assignment $\bar{x}$ from the master:

- **Step 1.** Compute realized workload in each scenario:

$$L_j^\omega(\bar{x}) = \sum_{i \in \mathcal{I}} S_{ij}^\omega \bar{x}_{ij}, \quad \forall j, \forall \omega.$$

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- **Step 2.** Determine scenario violations:

$$\delta_j^\omega(\bar{x}) = \begin{cases} 1, & \text{if } L_j^\omega(\bar{x}) > C_j, \\ 0, & \text{otherwise.} \end{cases}$$

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$$\delta_j^\omega(\bar{x}) = \begin{cases} 1, & \text{if } L_j^\omega(\bar{x}) > C_j, \\ 0, & \text{otherwise.} \end{cases}$$

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Feasibility Evaluation Subproblem

Given a candidate assignment $\bar{x}$ from the master:

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This is a **logical feasibility check**, not another optimization problem. Once $\bar{x}$ is fixed, all quantities are computed directly.

Feasibility Cut Generation

If agent j violates the chance constraint under $\bar{x}$:

- Let $\mathcal{I}_j(\bar{x}) = \{i \in \mathcal{I} : \bar{x}_{ij} = 1\}$ be the tasks assigned to agent j .

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- Then add the tighter cut:

$$\sum_{i \in \mathcal{B}_j(\bar{x})} x_{ij} \leq |\mathcal{B}_j(\bar{x})| - 1.$$

This excludes the specific offending combination while leaving more flexibility for the master.

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Convergence

The algorithm terminates in finitely many iterations because each feasibility cut eliminates at least one infeasible assignment pattern, and the number of binary assignment patterns is finite.

Why This Decomposition Works

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Why This Decomposition Works

- **Exact objective in master:** every master solution is evaluated with respect to the exact return-balance trade-off.
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- **Simple subproblem:** the feasibility check is purely arithmetic (no LP or MIP to solve).
- **Managerial interpretation:** the planner proposes an assignment, then checks whether it is sufficiently safe under workload uncertainty.

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Test Instances

Three instances with increasing size:

Instance	Tasks	Agents	Scenarios	α	λ
Small	5	3	3	0.20	4.0
Medium	10	5	5	0.20	5.0
Large	25	10	10	0.10	6.0

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Three instances with increasing size:

Instance	Tasks	Agents	Scenarios	α	λ
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Medium	10	5	5	0.20	5.0
Large	25	10	10	0.10	6.0

- Capacities vary slightly across agents (e.g., [10, 10, 11] for the small instance).
- The large instance uses a tighter risk tolerance ($\alpha = 0.10$).

Computational Summary

Instance	Objective	Return	Z	Iterations	Cuts	Gap
Small	51.50	53	0.376	2	1	0.00
Medium	77.11	79	0.377	52	220	0.00
Large	254.34	257	0.444	443	1958	0.00

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- All instances close the Benders gap to zero.
- The number of iterations and cuts increases substantially with instance size: from 2 iterations / 1 cut (small) to 443 iterations / 1958 cuts (large).

Small Instance: Assignment and Utilization

Assignment

Task	Agent
1	Agent 3
2	Agent 2
3	Agent 3
4	Agent 2
5	Agent 1

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Expected utilization

Agent	U_j
1	0.570
2	0.940
3	0.946

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Expected utilization

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3	0.946

- Solved in 2 iterations with 1 feasibility cut.
- The first master solution was infeasible because agent 1 received an unsafe bundle. One cut corrected this.

Medium Instance: Assignment and Utilization

Assignment

Task	Agent
2	Agent 2
3	Agent 3
4	Agent 4
5	Agent 5
7	Agent 1
10	Agent 4

Tasks 1, 6, 8, 9: **not assigned**.

Medium Instance: Assignment and Utilization

Assignment

Task	Agent
2	Agent 2
3	Agent 3
4	Agent 4
5	Agent 5
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10	Agent 4

Expected utilization

Agent	U_j
1	0.545
2	0.628
3	0.545
4	0.922
5	0.645

Tasks 1, 6, 8, 9: **not assigned**.

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Task	Agent
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Tasks 1, 6, 8, 9: **not assigned**.

- 52 iterations and 220 feasibility cuts.
- Leaving some tasks unassigned is optimal: assigning them would violate chance constraints or create too much imbalance.

Large Instance: Assignment and Utilization

Assignment highlights

- 19 of 25 tasks are assigned.
- Tasks 5, 11, 12, 17, 23, 24: not assigned.
- Most agents receive 2 tasks; agent 7 receives only 1.

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Agent	U_j
1	0.786
2	0.722
3	0.730
4	0.722
5	0.739
6	0.747
7	0.355
8	0.748
9	0.785
10	0.755

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443 iterations and 1958 cuts. Feasible but not perfectly balanced. Agent 7 is the lightest outlier.

Observations from Results

- **Scalability:** iterations and cuts grow substantially with instance size (from 2 to 443 iterations).

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- **Zero Benders gap**: all instances are solved to proven optimality, confirming correctness of the decomposition.
- **Computational cost**: the explicit outer-loop Benders scheme becomes expensive for large instances with tight chance constraints, suggesting room for acceleration strategies.

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Key Takeaways

- The formulation models cognitive resource allocation as a **stochastic multiple knapsack** with scenario-based chance constraints and a workload-balance penalty.

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- The formulation models cognitive resource allocation as a **stochastic multiple knapsack** with scenario-based chance constraints and a workload-balance penalty.
- **Benders decomposition** separates assignment decisions from chance-feasibility checks, keeping the exact objective in the master and using a simple logical subproblem.
- Computational results on three instances demonstrate correctness (zero gap) and reveal the scalability challenge as instance size grows.
- Future directions include stronger cut generation, scenario dominance [Büyüктаhtakın, 2023], and ML-accelerated decomposition [Larsen et al., 2024].

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